

The point model in Fortran: coupling without the material server

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Why

In the coupled runs the field of Klempt et al. (2024) sets the amount of biofilm ϕ and the point model of Klempt et al. (2026) sets its composition at every integration point. Until now the point model ran in Python: the material routine sent the state of each integration point to a material server through a socket and waited for the answer. This call took about 100 times as long as the rest of a run. Two species on 2880 elements needed 1.5 h in Abaqus, one species on 40^3 elements without the server 16 min.

I therefore wrote the point model in Fortran (`ecology_native.f`). It replaces the Python bridge in the build and keeps its interface, so the material routines of ANSYS and Abaqus use it without any change. No server, no socket and no C layer are needed.

What it computes

The state of one point is $\mathbf{g} = (\phi_1, \dots, \phi_5, \phi_0, \psi_1, \dots, \psi_5, \gamma)$, with $n \leq 5$ active species (unused species stay exactly zero). One coupling step of length $s\Delta t$ is divided into sub-steps of at most 10^{-4} . In each sub-step the twelve residuals $\mathbf{R}(\mathbf{g}_{k+1}; \mathbf{g}_k) = \mathbf{0}$ of the point model (Klempt et al. 2026, Eq. 16–18, divided by η_i) are solved by six Newton iterations,

$$\mathbf{g} \leftarrow \mathbf{g} - \left(\frac{\partial \mathbf{R}}{\partial \mathbf{g}} \right)^{-1} \mathbf{R},$$

with the state clipped to its admissible range before and after each iteration. The Jacobian is the exact derivative, written out by hand; the linear system is solved by LU decomposition with partial pivoting. This is the same scheme as the Python reference, which computes the Jacobian by automatic differentiation.

The constants of a case (n , c^* , α^* , η_i and the interaction matrix) are read once from a small text file, written by `write_eco_cfg.py <case>` from the same case table as the Python server. As in the server, the interaction matrix in the input deck is checked against the case. The local nutrient enters as $c^* = c_0^* \min(\max(c/c_{\text{ref}}, 0), 1)$.

Checks

Check	Result
Single points against the Python reference: 16 states, two-species cases 3 and 6, four-species case 1 and the five-species model, several sub-step counts and nutrient levels (<code>tests/test_ecology_native.py</code>)	largest relative difference 8×10^{-15}
ANSYS, the partner's element, 8^3 elements, two species, same deck as a run with the Python server	stresses and composition identical in every printed digit; trace within 1.6×10^{-12}
Abaqus, 8^3 elements, two species with the nutrient, same deck as a run with the Python server	printed values identical; trace of 8213 point steps within 6.7×10^{-12}

The remaining differences are round-off: the Fortran Jacobian and the automatically differentiated one sum their terms in a different order.

Speed

Run	Python server	Fortran
ANSYS, partner element, 8^3 , two species	156 s	36 s
Abaqus, 8^3 , two species with the nutrient	5.2 min (4 CPUs)	0.6 min (1 CPU)
Abaqus, 16^3 , two species	10 min for 30 % of the run	1 min (whole run)

The point model now costs about as much as the mechanics. Runs that were out of reach with the server, such as two species on 32^3 elements, take hours instead of days.

How to use it

- ANSYS: build the custom executable with `ecology_native.f` in place of the Python bridge (`usermat_py_hook.f`); the material routine and the call-site code stay as they are. Before starting ANSYS, set the environment variable `BIOFILM_ECO_CASE` to the case file. One process per run.
- Abaqus: `run_comp.ps1 -Native` builds and links the same file into the user subroutine; the case file is read once at the start of the analysis.
- The number of species in the material input must match the case file, as it must match the server's case.

Limits

- Cases with a nutrient level or antibiotic level that changes in time (four-species cases 3 and 4 of Klempt et al. 2026) are not supported; the Python server does not support them either.
- The Hill gate of this repository's five-species model is off, as in all current runs.
- The number of Newton iterations is fixed at six, as in the reference. Where a sub-step is too long for six iterations to converge, both versions end at the clip bounds and can differ there; the sub-steps of the composition runs converge.

Source: `ansys_usermat/coupling/ecology_native.f`; checks and timings:
`abaqus_composition/README.md`; repository `keisuke58/pde-fem-biofilm`.